

Exercises in Marine Ecological Genetics

04. Genome sequencing and assembly

- Short- and long-read sequencing
- Raw sequencing data
- Genome assembly
- Command-line basics

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<https://github.com/mhelmkampf/meg26>



Image by Jake Wilton / Ocean Image Bank

Course outline

Preliminary, may be subject to change

Class	Date	Topics	Script
01	Apr 10	Introduction and setup	01_intro.R
02	Apr 17	Hardy-Weinberg equilibrium	02_hwe.R
03	Apr 24	<i>F</i> -statistics and population structure I	03_popst.R
–	May 01	Labor Day	
04	May 08	Genome sequencing and assembly	04_asm.sh
–	May 15	Himmelfahrt Break	
05	May 22	Variant calling and SNPs	05_vcal.sh
06	May 29	Population genomics and genetic diversity	06_gdiv.sh
07	Jun 05	Population structure II	07_clust.sh
08	Jun 12	Selection	08_sel.R
–	Jun 19	Student presentations (no exercises)	
09	Jun 26	DNA barcoding	09_barcode.sh
10	Jul 03	Metabarcoding / eDNA	10_microb.txt
11	Jul 10	Metabarcoding / eDNA II (optional)	11_edna.R

F_{IS} and self-fertilization in hamlets

Homework

Hamlets are simultaneous hermaphrodites with external fertilization. Discuss whether self-fertilization occurs regularly according to F -statistics.

- How would the number of heterozygotes be affected by self-fertilization?
- What would you expect for the inbreeding coefficient F_{IS} in this case?
- Use the *H. puella* dataset ("caribbean") to check your prediction



Photograph by Luiz Rocha

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F_{IS} and self-fertilization in hamlets

Homework

Hamlets are simultaneous hermaphrodites with external fertilization. Discuss whether self-fertilization occurs regularly according to F -statistics.

```
### Calculate global F-statistics
genepop::Fst("data/msats/puella_caribbean.gen",
            outputFile = "work/Fst_caribbean.txt")

Fwc(is) All: 0.0023 # see bottom of output file
```

- Use the *H. puella* dataset ("caribbean") to check your prediction
- F_{IS} is very low, **no evidence for self-fertilization**

Today's objectives

By the end of this session, you will be able to:

- Compare **major genome sequencing technologies** and their key trade-offs
- Inspect and process raw sequencing data in **Fastq format**
- Generate a **draft genome assembly** and assess assembly quality using contiguity and completeness metrics
- Use **basic bash commands** to navigate files and execute simple assembly steps

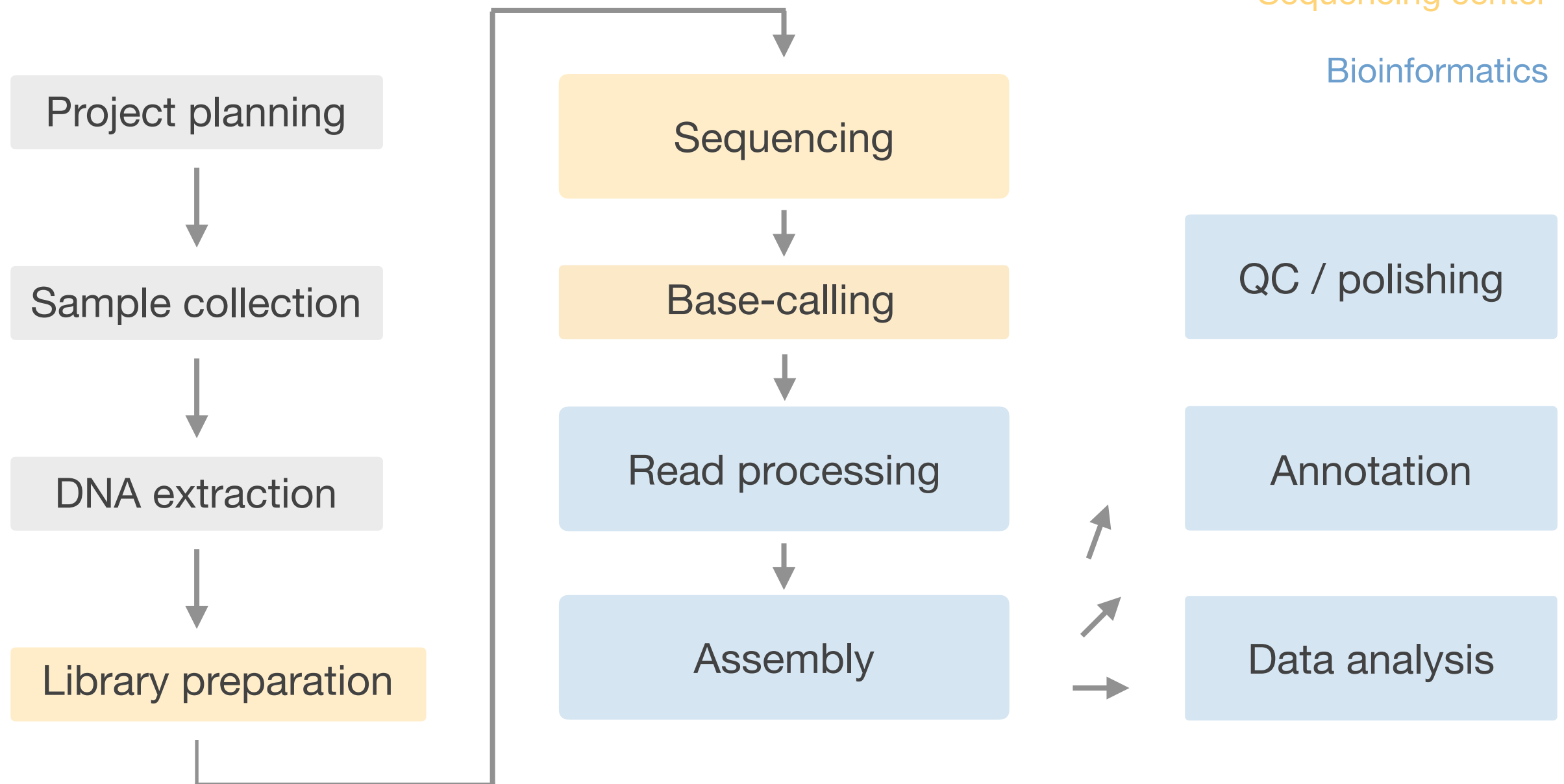
De novo genome sequencing workflow

Legend

Preparation

Sequencing center

Bioinformatics



Exercise 1

```
head -n 4 HypPue1_illumina_raw_F.fastq # display first 4 lines of file
```

```
@HWI-ST1293:199:HA9JHADXX:1:1101:1044:1603 1:N:0:CGATGT  
NCCCTGTTAAAGGATCATCTCTGACCTATCATTGTGGTGTAAATCACATTTAACACAATCACGATGTGCTTTACCTGCAGC  
ATCTTTACAGCAGGGCTGGGAGATATGACC AAAACAGTTATGATAATATGTTTTATTTCTATTGAAAATCA  
+  
#1=DDFFHHHHHHJJJJJJJJJJJJJJJJJJJJJJGHJJJJJJIIJJJJHJJJJJJJJJJJJJJJJJJJJHHHHH  
FFFFFFEEEEEDDDDDDDDD@DDDEDEDDBDDDDDDDCDCEDDEEDEEEDEECDDDEDEDEEDDDDDDDDC
```

1. @ followed by sequence id and optional info (e.g. instrument/run id, barcode)
2. DNA sequence
3. +, sometimes followed by sequence id
4. base quality score (same length as sequence)

Base quality

Phred **quality score**:

$$Q = -10 \log_{10} P$$

Common benchmark:

% bases with $Q \geq 30$

Quality score	<i>P</i> incorrect base call	Base call accuracy
10	1 in 10	90 %
20	1 in 100	99 %
30	1 in 1000	99,9 %
40	1 in 10000	99,99 %

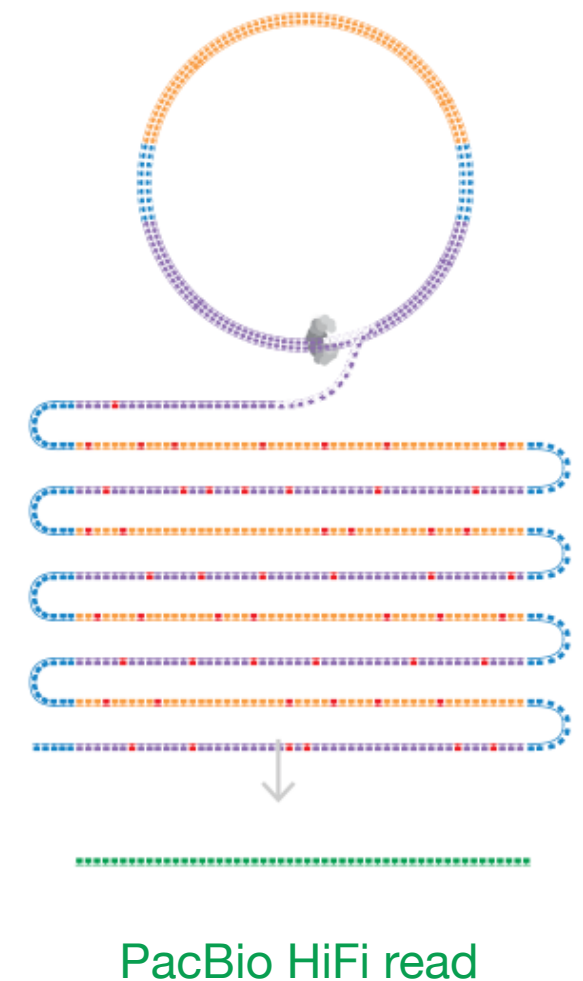
FASTQ encoding (Illumina 1.8+):

ASCII Symbol:	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJ
Quality Score:	0.2.....10.....20.....30.....41

Genome assembly

Exercise 2

- Reconstructing long, continuous sequence from millions of overlapping **reads**
- Reads can be very short (e.g. Illumina) or long (e.g. PacBio, ONT)
- Segments of assembled sequence are called **contigs**, which may be combined into scaffolds
- Scaffolds or PacBio contigs can be up to chromosome-length



Sequencing technologies compared

Exercise 2

Technology	Read length	Accuracy	Gb per run	Human genome	Devices
Illumina	2 × 150 bp	> 99.9 %	1000–10000	300–700 €	NextSeq 2000, NovaSeq X
PacBio	10–20 kb	> 99.8 %	80–300	1200–2500 € ↓	Sequel II, Revio
ONT	10–100 kb	97–99 % ↑	50–200	900–2200 € ↓	MinION, PromethION

As of 2026; typical values in real-world academic settings; ↓ ↑ trends

bp = base pairs, kb = kilo bases (1000 bp), Mb = Mega bases (million bp), Gb = Giga bases (billion bp)

Human genome: cost for 30 × coverage, including library prep (Sonnet 4.6 estimate)

Sequencing technologies compared

PacBio

Illumina

- High yield, low cost, most accurate
- Short reads limit contiguity
- ▶ Whole genome resequencing (population genomics), eDNA / metabarcoding

- Long reads, accurate
- Expensive, requires high-quality DNA
- ▶ High-quality reference genome assemblies, structural variation, phasing

ONT

- Portable, longest reads possible
- Less accurate
- ▶ eDNA / metabarcoding, projects in low-resource settings



Photo: Achim Meyer

Assessing assembly quality

Exercise 2

- Sequencing depth / coverage
- Assembly metrics: size distribution of contigs / scaffolds
- Average base accuracy (Q score)
- Percentage of assembly assigned to chromosomes
- Gene completeness
- Phasing information

Challenges

- Contamination
- Misassembled regions
- Presence of false duplications

Sequencing depth / coverage

Exercise 2

- Average number of reads representing each position in the genome
- coverage or depth = $\text{read count} \times \text{average read length} / \text{genome size}$
- high coverage facilitates assembly, detection of sequencing errors
- Typical coverage: 50–100 × (or more) for *de novo* genome sequencing

10–30 × for re-sequencing

Genome: CGTAATGGCATATCGCCTAGATTTCGAAACG

Read 1: TAATGGCATATCGCCTAGAT

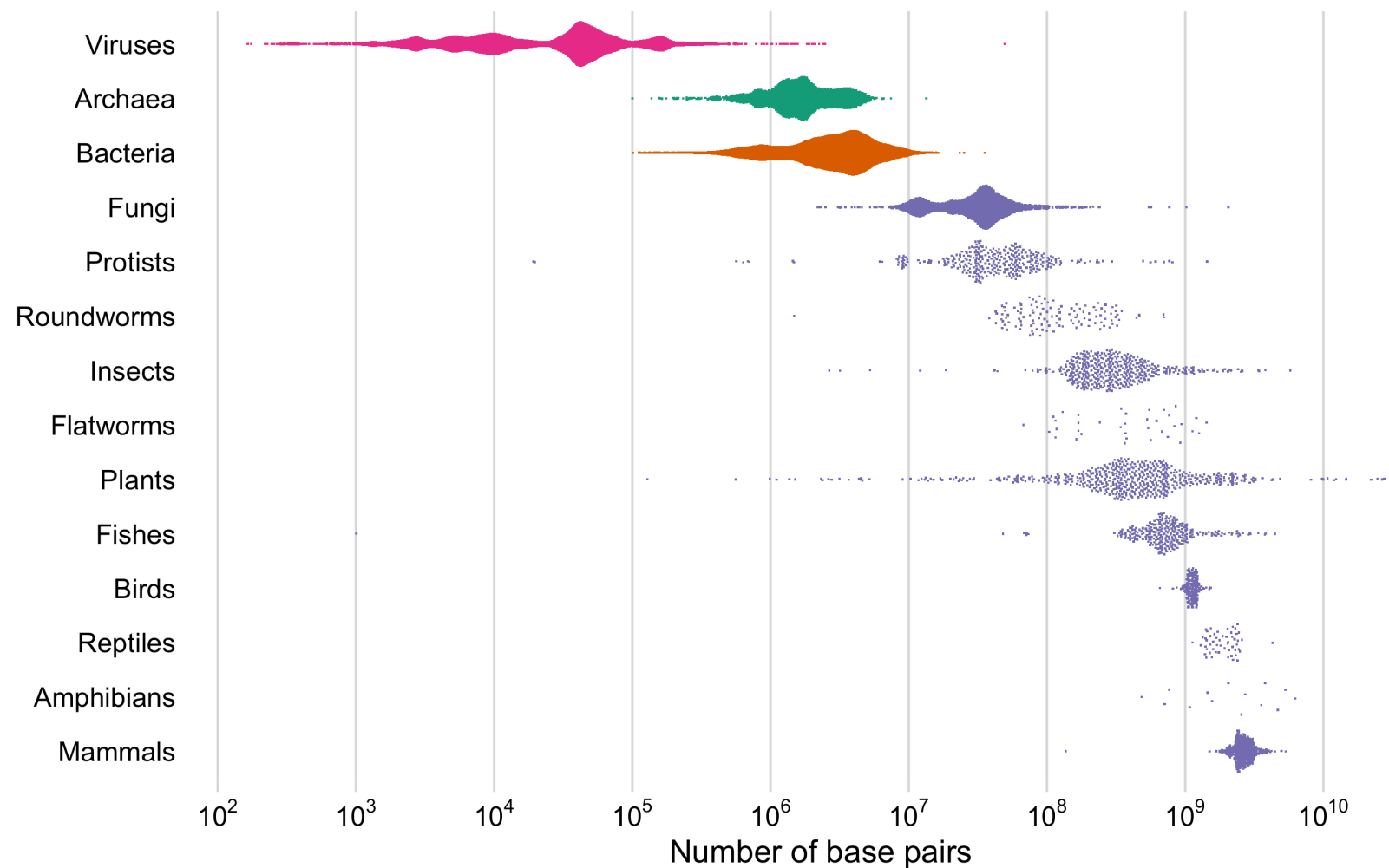
Read 2: CATATCGCCTAGATTTCGAAA

Read 3: TATCGCCTAGATTTCGAAACG

Depth: 00111111223333333333333322222211

Genome size

Exercise 2



- Nine orders of magnitude
- Genome size correlated with repetitive DNA, which is hard to sequence and assemble

Based on 50,000 organisms with genome information in NCBI

See also www.genomesize.com

Tom E. White, Datavision 2020

- Total size (compare to expected genome size)
- Number of contigs / scaffolds
- Largest scaffold
- **N50**: contig / scaffold size where 50% of assembly is found on contigs / scaffolds of equal or larger size (measure for fragmentation / sequence contiguity)

Scaffolds: 530, 760, 1050, 610, 450, 800, 220, and 1200 kb

Reorder: 1200, 1050, 800, 760, 610, 530, 450, 220 kb

Sum/2: $5620/2 = 2810$

Add up until sum/2 is reached: $1200 + 1050 + 800 > 2810$

N50 = 800 kb

The Blackchin guitarfish genome

Exercise 2/3

PacBio HiFi sequencing of *Glaucostegus cemiculus*

Assembly using hifiasm

Kirsti Burnett

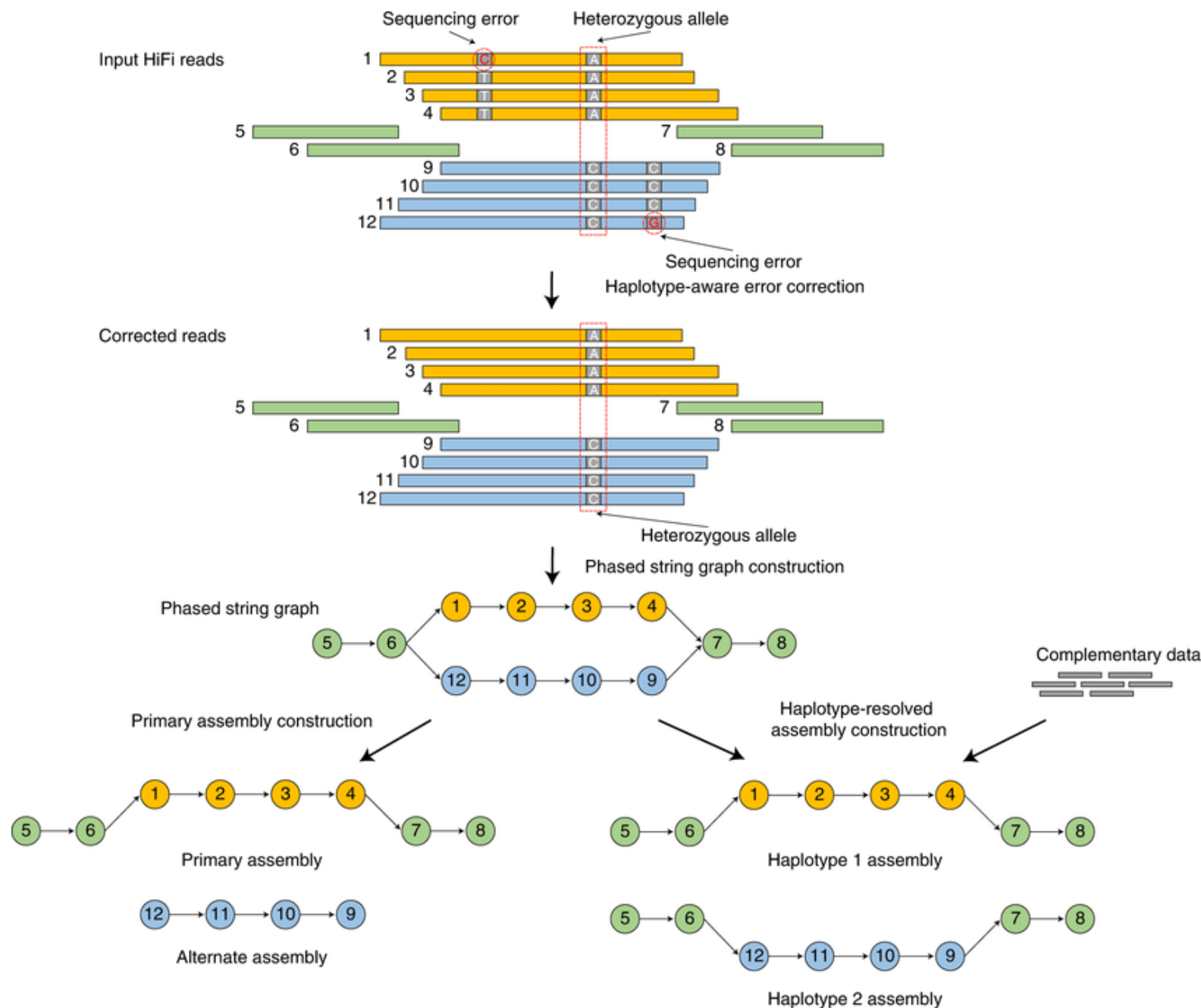


Genome assembly stats

Size	2.85 Gb
No. contigs	1511
N50	11.9 Mb
Coverage	18 ×

Primary vs. haplotype assemblies (hifiasm)

Exercise 3

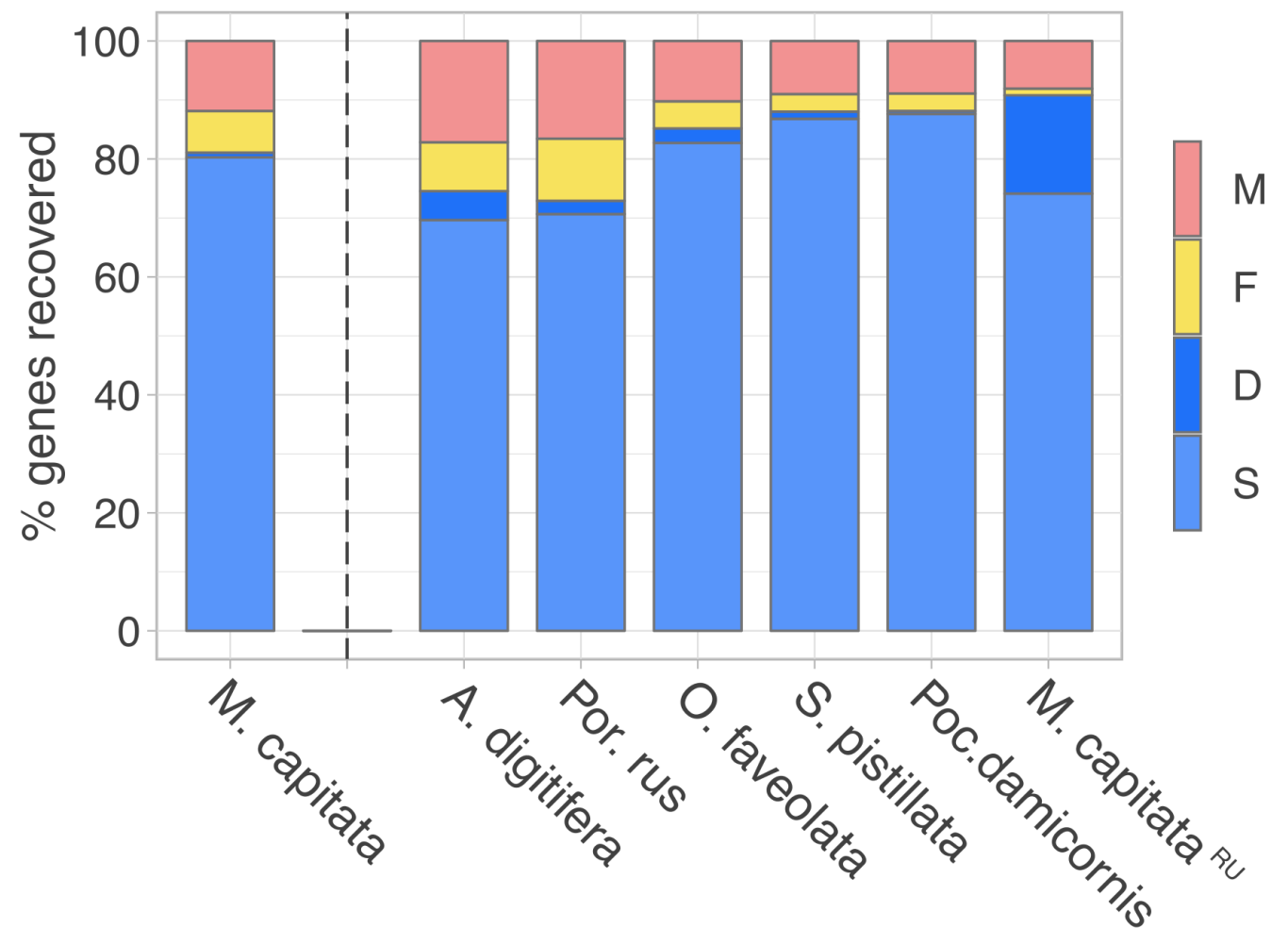


Cheng et al. 2021 *Nature Methods*

Gene completeness with BUSCO

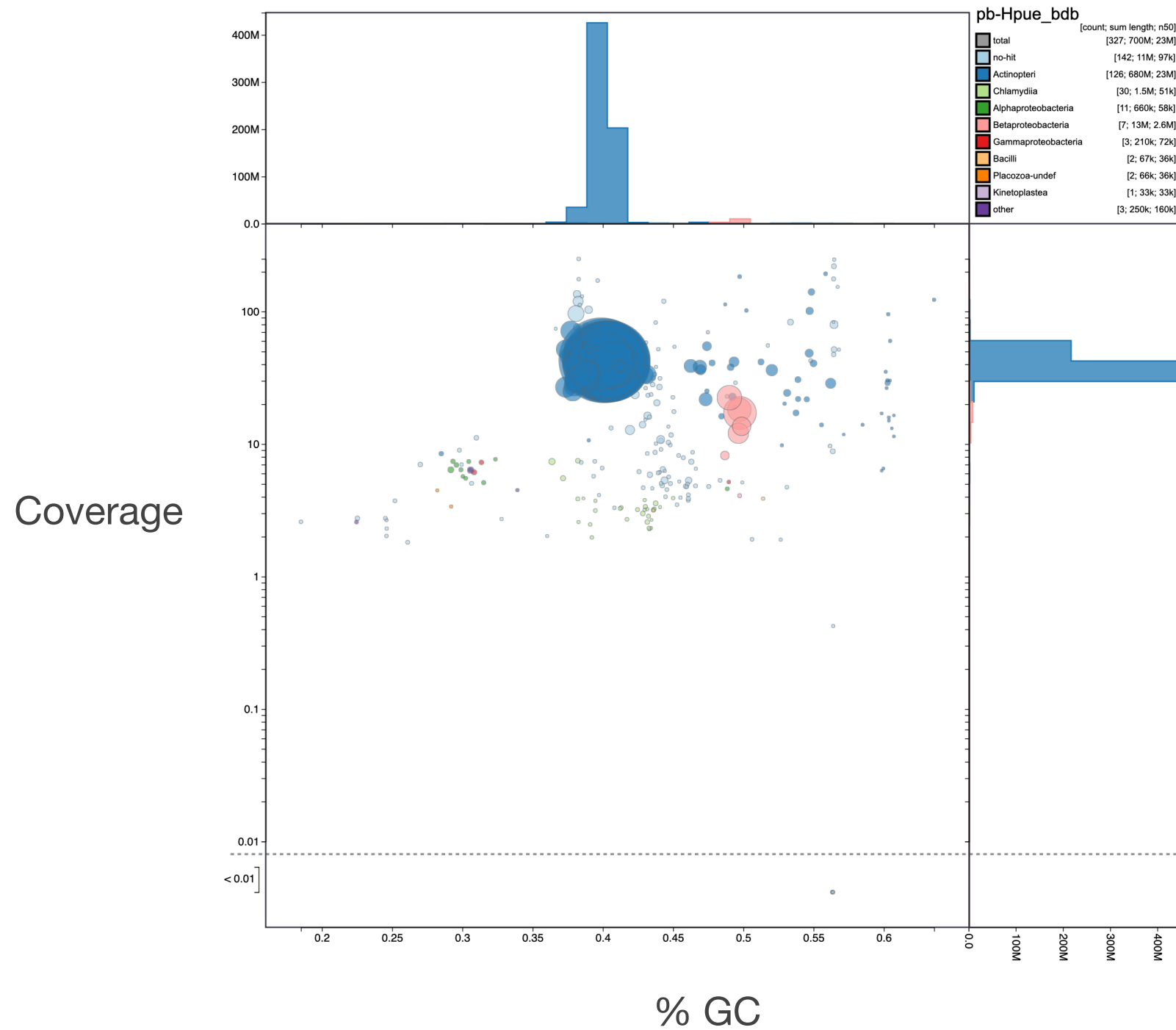
<https://busco.ezlab.org>

Quantifies assembly completeness based on presence of universal, highly conserved, single-copy genes (e.g. housekeeping genes)



Helmkamp et al. 2019 (Genome Biology and Evolution)

Contamination QC with BlobTools



Color:
Most similar
known taxon

HypPue2.1_pacbio_pctg.fas

Basic bash commands

Summary

<code>ls -l</code>	# List directory contents in long format
<code>cd <directory></code>	# Change into directory
<code>cd ..</code>	# Change into directory above / back
<code>mkdir <directory></code>	# Create directory
<code>mv <file/dir> <file/dir></code>	# Move or rename file or directory
<code>cat <file></code>	# Print file to screen
<code>head -n 100 <file></code>	# Print first 100 lines of file

tab key	# Autocomplete file or directory name
up arrow key	# Cycle through previous commands

Genome sequencing and assembly

Summary

```
grep -c '^+$' <raw_reads>.fastq      # Calculate number of reads in Fastq file  
  
assembly_stats <assembly_file>.fas    # Calculate assembly metrics (Python tool)  
  
hifiasm -o <output_dir> --primary <read_file>.fastq    # Assemble PacBio reads
```

- **Short- and long-read sequencing** technologies offer different but complementary strengths
- **Assembly quality** can be evaluated using metrics like coverage, contiguity (e.g. N50) and completeness (e.g. BUSCO)
- **Assemblies are drafts** – often fragmented and with errors – but serve as the foundation for further analyses
- Genome-scale data have **many applications**, including generating large SNP datasets for population genetic analyses, inferring population history, and detecting genes under selection